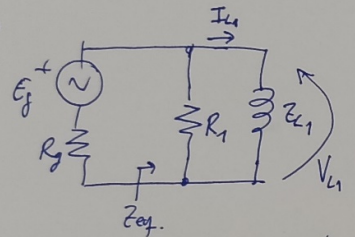
P1

a) Resolvemos el circuito antes de la conmutación, para $t<0$, en el cual se encuentra conectado al generador $e_g(t) = 4\cos(10^3 t)\,V$ desde $t=-\infty$, esto es, el circuito está en régimen permanente sinusoidal $\Rightarrow$ resolvemos usando fasores e impedancias

$t<0$)



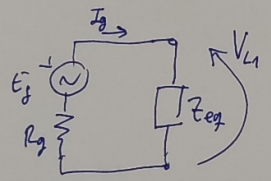
$\dot{E}_g = 4V$, $\omega = 10^3\ rad/s$

$Z_{L_1} = j\omega L_1 = j\,k\Omega$

Resolvemos simplificando mediante la asociación de R_1 y Z_{L_1}:

$$Z_{eq} = R_1 // Z_{L_1} = \frac{2+4j}{5}\,k\Omega$$

$$I_g = \frac{\dot{E}_g}{R_g + Z_{eq}} = \frac{3-j}{2}\,mA$$



$$V_{L_1} = Z_{eq}\cdot I_g = (1+j)\,V$$

$$I_{L_1} = \frac{V_{L_1}}{Z_{L_1}} = (1-j)\,mA = \sqrt{2}\,e^{-j\frac{\pi}{2}}\,mA$$

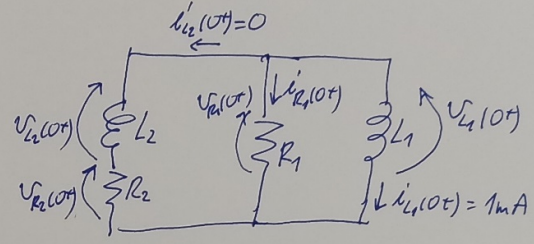
La expresión temporal: $i_{L_1}(t) = \sqrt{2}\cos\left(10^3 t - \frac{\pi}{2}\right)mA$

En el instante justo después de la conmutación ($t=0^+$) es igual al valor de la corriente a través de L justo antes ($t=0^-$)

$$\boxed{i_{L_1}(0^+) = i_{L_1}(0^-) = \sqrt{2}\cdot\frac{1}{\sqrt{2}}\,mA = 1mA}$$

P1

b) En $t=0^+$, justo después de la conmutación: $\begin{cases} i_{L_1}(0^+) = i_{L_1}(0^-) = 1mA \\ i_{L_2}(0^+) = i_{L_2}(0^-) = 0 \end{cases}$



$$v_{L_1}(0^+) = v_{R_1}(0^+)$$

$$v_{R_1}(0^+) = R_1\cdot i_{R_1}(0^+)$$

$$i_{R_1}(0^+) = -i_{L_1}(0^+) = -1mA$$

$$\Rightarrow v_{R_1}(0^+) = 2\,k\Omega\,(-1mA) = -2V$$

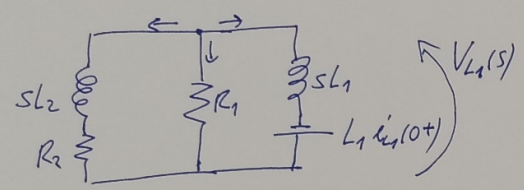
$$\Rightarrow \boxed{v_{L_1}(0^+) = -2V}$$

$$v_{L_2}(0^+) + v_{R_2}(0^+) = v_{L_1}(0^+) \Rightarrow v_{L_2}(0^+) = v_{L_1}(0^+) - v_{R_2}(0^+)$$

como $i_{L_2}(0^+) = 0 \Rightarrow v_{R_2}(0^+) = 0$

$$\Rightarrow \boxed{v_{L_2}(0^+) = -2V}$$

c) Para $t>0$



$$\frac{V_{L_1}(s)}{R_2+sL_2}+\frac{V_{L_1}(s)}{R_1}+\frac{V_{L_1}(s)+L_1 i_{L_1}(0^+)}{s\cdot L_1}=0$$

$$V_{L_1}(s)\left[\frac{1}{R_2+sL_2}+\frac{1}{R_1}+\frac{1}{sL_1}\right]=\frac{-L_1 i_{L_1}(0^+)}{s\cdot L_1}$$

$$\Rightarrow V_{L_1}(s)=\frac{-\frac{L_1 i_{L_1}(0^+)}{s\cdot L_1}}{\frac{1}{R_2+sL_2}+\frac{1}{R_1}+\frac{1}{sL_1}}=\frac{-i_{L_1}(0^+)}{\frac{s}{R_2+sL_2}+\frac{1}{R_1}s+\frac{1}{L_1}}$$

intentamos conseguir el polin. caracteristico en el denominador para ello multiplicamos numerador y denominador $\times(R_2+sL_2)$

$$\Rightarrow V_{L_1}(s)=\frac{-i_{L_1}(0^+)(R_2+sL_2)}{s+\frac{1}{R_1}(R_2+sL_2)\cdot s+\frac{1}{L_1}(R_2+sL_2)}$$

$$=\frac{-i_{L_1}(0^+)(R_2+s\cdot L_2)}{\frac{L_2}{R_1}s^2+\left(1+\frac{R_2}{R_1}+\frac{L_2}{L_1}\right)s+\frac{R_2}{L_1}}\quad\left|\times\frac{R_1}{L_2}\right.\quad\left|\times\frac{R_1}{L_2}\right.$$

$$\boxed{V_{L_1}(s)=\frac{-i_{L_1}(0^+)R_1\left(\frac{R_2}{L_2}+s\right)}{s^2+\left(\frac{R_1}{L_2}+\frac{R_2}{L_2}+\frac{R_1}{L_1}\right)s+\frac{R_2\cdot R_1}{L_1\cdot L_2}}}$$

P1

d) La ecu. caracteristica

$$s^2+\left(\frac{R_1}{L_2}+\frac{R_2}{L_2}+\frac{R_1}{L_1}\right)s+\frac{R_2R_1}{L_1L_2}=0$$

queda

$$s^2+\underbrace{5000}_{2\zeta\omega_n}s+\underbrace{4\cdot10^6}_{\omega_n^2}=0$$

$$\Rightarrow \boxed{\omega_n=\sqrt{4\cdot10^6}=2k\ \text{rad/s}}$$

$$2\zeta\omega_n=5000\Rightarrow\boxed{\zeta=\frac{5000}{2\omega_n}=\frac{5}{4}=1{,}25\angle 1}$$

$\Rightarrow$ El circuito es sobreamortiguado

e) Las raices del polin caracteristico resultan.

$$s_1=-1000,\ s_2=-4000$$

$$\Rightarrow V_{L_1}(s)=\frac{-2(2000+s)}{(s+1000)(s+4000)}\overset{\text{desarrollo fracciones simples}}{=}\frac{A}{s+1000}+\frac{B}{s+4000}$$

$$\Rightarrow -2(2000+s)=A(s+4000)+B(s+1000)$$

para $s=-4000\Rightarrow 4000=B\cdot(-3000)\Rightarrow B=-\frac{4}{3}$

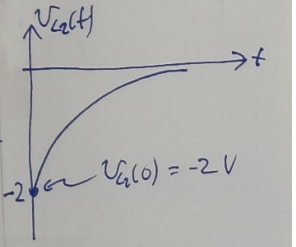
" $s=-1000\Rightarrow -2000=A\cdot3000\Rightarrow A=-\frac{2}{3}$

$$\Rightarrow V_{L_1}(s)=\frac{-\frac{2}{3}}{s+1000}+\frac{-\frac{4}{3}}{s+4000}$$

La transformada de Laplace

$$\boxed{v_{L_1}(t)=\left(-\frac{2}{3}e^{-1000t}-\frac{4}{3}e^{-4000t}\right)u(t)\,V}$$

$$=-\frac{2}{3}\left(e^{-1000t}-2e^{-4000t}\right)u(t)\cdot V$$



REDMI NOTE 8 PRO

P2) $V_1 = h_{11} I_1 + h_{12} V_2$
$I_2 = h_{21} I_1 + h_{22} V_2$ $\Big\}$ $V_2 = 0 \Rightarrow$ $V_1 = h_{11} I_1 \rightarrow h_{11} = \frac{V_1}{I_1}\Big|_{V_2=0}$
$I_2 = h_{21} I_1 \rightarrow h_{21} = \frac{I_2}{I_1}\Big|_{V_2=0}$

$I_1 = \frac{E_g}{R_g + Z_{MN}} = \frac{4}{1+1-2j} = 1+j \text{ A}.$

$V_1 = Z_{MN} \cdot I_1 = (1-2j)\cdot(1+j) = 3-j \text{ V}.$

$h_{11} = \frac{V_1}{I_1}\Big|_{V_2=0} = \frac{3-j}{1+j} = \frac{(3-j)(1-j)}{2} = \frac{3-3j-j+j^2}{2} = \frac{2-4j}{2} = 1-2j \ \Omega$

$h_{21} = \frac{I_2}{I_1}\Big|_{V_2=0} = \frac{1+j}{1+j} = 1$

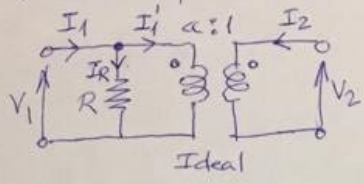
SIMÉTRICO $\begin{cases} h_{21} = -h_{12} \rightarrow h_{12} = -h_{21} = -1 \\ |h| = 1 \end{cases}$

$h_{11} h_{22} - h_{12} \cdot h_{21} = 1 \rightarrow h_{22} = \frac{1 + h_{12} h_{21}}{h_{11}} = 0$

$$h_Q = \begin{pmatrix} 1-2j\ \Omega & -1 \\ 1 & 0 \end{pmatrix}$$

P3) Conexión paralelo-serie ⇒ parámetros "g"

$$\begin{cases} I_1 = g_{11} V_1 + g_{12} I_2 \\ V_2 = g_{21} V_1 + g_{22} I_2 \end{cases}$$



Ec's transformador

$$\begin{cases} I_1' = -\frac{1}{a} I_2 = -\frac{1}{2} I_2 \\ V_2 = \frac{1}{a} V_1 = \frac{1}{2} V_1 \end{cases}$$

$$I_1 = I_R + I_1' = \frac{V_1}{R} + I_1' \Rightarrow I_1 = \frac{1}{R} V_1 - \frac{1}{a} I_2 = \frac{1}{R} V_1 - \frac{1}{2} I_2$$

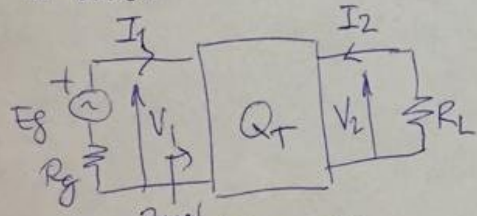
Por tanto, las ec's en parámetros "g" del cuadripolo:

$$\begin{cases} I_1 = \frac{1}{R} V_1 - \frac{1}{2} I_2 \\ V_2 = \frac{1}{2} V_1 \end{cases} \quad \begin{matrix} g_{11} = \frac{1}{R} & g_{12} = -\frac{1}{2} \\ g_{21} = \frac{1}{2} & g_{22} = 0 \end{matrix}$$

El cuadripolo total:

$$(g_{QT}) = \begin{pmatrix} \frac{1}{R} & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{R}+\frac{1}{2} & -1 \\ 1 & 0 \end{pmatrix}$$

Por tanto:



$$I_1 = \left(\frac{1}{R}+\frac{1}{2}\right) V_1 - I_2 \quad \begin{cases} (1) \\ (2) \end{cases} \text{EC's PARÁMETRO "g"}$$

$$V_2 = V_1$$

$$V_2 = -R_L I_2 = -2 I_2 \quad (3) \text{ EC. PUERTA SALIDA.}$$

Sustituyendo (3) en (2):

$$V_1 = -2 I_2 \longrightarrow I_2 = -\frac{1}{2} V_1$$

Y sustituyendo en (1):

$$I_1 = \left(\frac{1}{R}+\frac{1}{2}\right) V_1 + \frac{1}{2} V_1 = \left(\frac{1}{R}+\frac{1}{2}+\frac{1}{2}\right) V_1 = \left(\frac{1}{R}+1\right) V_1$$

$$Z_{MN} = \frac{V_1}{I_1} = \frac{1}{\left(\frac{1}{R}+1\right)} = \frac{R}{R+1} = R_g = \frac{1}{3}\,\Omega$$

$$\frac{R}{R+1} = \frac{1}{3} \rightarrow 3R = R+1 \rightarrow 2R = 1 \rightarrow \boxed{R = \frac{1}{2}\,\Omega}$$